

CO2-AT1

Name : M. Hema Kumar

Reg. NO : 192521138

Course code : CSE0302

Q1. Find - Largest using Divide and Conquer :

= 1) Introduction :

* Divide and conquer is an algorithm design technique.

* It divides the problem into smaller sub problems, solves them recursively and combines the results to obtain the final solution.

2) Algorithm :

= Step 1 : Start

Step 2 : If low = high
Return A [low]

Step 3 : mid = $(\text{low} + \text{high}) / 2$

Step 4 : Leftmax = Find - Largest (A, low, mid)

Step 5 : Right max = Find - largest (A, mid + 1, high)

Step 6 : If Left max > Right max

Return Leftmax

Else

Return Right Max

Step 7 : Stop.

3. Example :

Array :

$$A = [15, 8, 22, 6, 30, 12, 18, 9]$$

Result :

Left Half maximum = 22

Right Half maximum = 30

Largest element = 30.

4. Complexity :

Time complexity = $O(n)$

Space complexity = $O(\log(n))$

5. Advantages :

* Easy to solve recursively.

* Efficient for large problems.

* Forms the basis of many Divide-and-conquer algorithms.

Print Prime Numbers in a Singly Linked List :

Introduction :

- * A singly linked list is a linear data structure
- * Each node contains data and a pointer to the next node.
- * Traverse the list and print only the prime numbers.

Algorithm :

Step 1: start

Step 2: $ptr = head$

Step 3: while $ptr \neq NULL$

Step 4: Read $ptr \rightarrow data$

Step 5: If IsPrime (data)

Print &data

Step 6: $ptr = ptr \rightarrow next$

Step 7: stop

Prime checking :

If $n \leq 1 \rightarrow \text{Not Prime}$

If $n = 2 \rightarrow \text{Prime}$

If n is even $\rightarrow$ Not Prime

Otherwise check divisibility from 3 to $\sqrt{n}$.

If divisible $\rightarrow$ Not Prime

Else $\rightarrow$ Prime.

3. Example:

Linked List:

24 $\rightarrow$ 21 $\rightarrow$ 57 $\rightarrow$ 31 $\rightarrow$ 12 $\rightarrow$ 47 $\rightarrow$ 7 $\rightarrow$ 38 $\rightarrow$ NULL

Prime numbers:

31 $\rightarrow$ 47 $\rightarrow$ 7

4. Complexity:

* Time Complexity $= O(n\sqrt{n})$

* Space Complexity $= O(1)$

5. Advantages:

* Single traversal of the linked list.

* Uses constant memory.

* Simple and efficient.